Alan Anto

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Education

- **Brandenburg University of Technology** — Cottbus, Germany
  MSc in Artificial Intelligence, Grade: 2.2/5 — *Apr 2025 - Sep 2027*
  Courses: *Cognitive Systems, Image Processing, Neural Networks, Explainable Machine Learning*
- **APJ Abdul Kalam Technological University** — Kerala, India
  B.Tech (Hons.) in Artificial Intelligence and Data Science, Grade: 9.1/10 — *Dec 2020 - Jun 2024*
  Courses: *Algorithms and Data Structures, Artificial Intelligence, Machine Learning, NLP, Database systems*

Skills Summary

- **Languages:** Python, SQL, C++
- **GenAI Framework:** LLM, Transformers, RAG, Prompt Engineering, QLoRa, Hugging Face
- **Agentic AI Frameworks:** LangGraph, BeeAI, crewAI, NeMo, Google ADK, MCP, Guardrails
- **ML & Deep Learning:** PyTorch, TensorFlow, Keras, Scikit-learn, Pandas, Numpy
- **Databases:** MySQL, Redis, Chroma, FAISS(Vector), Neo4j(Graph), PostgreSQL
- **Cloud & DevOps:** AWS, Azure, Docker, Git, Gradio, REST API, WebSocket
- **Soft Skills:** Leadership, Documentation, MS Office, Time Management

Experience

- **Indian Institute of Information Technology** — Kerala, India
  AI Engineering Intern — *Jun 2023 - Aug 2023*
  - Achieved **30 FPS** real-time inference on resource-constrained edge hardware by fine-tuning **YOLOv8** (**PyTorch**) on **5,000+** annotated images and optimizing the deployment pipeline.
  - Reduced **model size** and **inference latency** while maintaining minimal accuracy loss by applying **PyTorch quantization** and exporting models to **ONNX** for optimized edge inference.
  - Improved detection robustness by **15%** across diverse real-world conditions by engineering a custom augmentation pipeline with **OpenCV** and **PyTorch transforms**.

Projects

- **AutoRAG Intelligence** | *Python, LangGraph, Neo4j, ChromaDB, FastAPI, Gradio, Ollama, Docker, Git* — LINK
  - Reduced hallucinations by **40%** and optimized routing efficiency by engineering a production-grade **hybrid RAG** system with a **10-node LangGraph agent pipeline** integrating **Neo4j** and **ChromaDB**.
  - Improved response delivery speed by **40%** by building a secure **full-stack** application with an **Async FastAPI** backend, Gradio frontend, and local LLM inference (Ollama), containerized with **Docker**.
  - Increased retrieval accuracy by **25%**, validated using **RAGAS**, by developing an end-to-end **data ingestion and indexing pipeline** with custom automotive **entity extraction** and a **hybrid Neo4j indexing** schema.
- **Autonomous Multi-Agent Travel System** | *Python, BeeAI, Git, FastAPI, React, Async, API Tools* — LINK
  - Achieved **100% automated data synthesis** by engineering a **multi-agent travel planner** with a hierarchical **BeeAI** architecture, specialized expert agents, and structured **handoff protocols**.
  - Improved LLM execution reliability by implementing **guardrails** with **BeeAI RequirementAgents**, enforcing conditional tool execution and state consistency across complex agent **handoffs**.
  - Reduced user-perceived latency by **65%** by developing an **asynchronous FastAPI** backend that streamed real-time agent trajectories and tool events via **SSE** to a **React** UI.
- **Agentic Code-Mate** | *Python, TypeScript, React, Google ADK, Groq, FastAPI, pytest, Llama, Git, Docker* — LINK
  - Accelerated developer workflows by **35%** by engineering a **multi-agent AI code editor** using **Google ADK** and **LLaMA 3.3 70B**, routing prompts across **7 specialist agents** through an intelligent Orchestrator.
  - Enabled real-time **token-by-token code streaming** by architecting a **FastAPI WebSocket** backend with a **TypeScript and React** frontend supporting syntax-highlighted and copyable code blocks.
  - Reduced deployment complexity to a **single command** by containerizing the full stack with **Docker Compose**, integrating a multi-stage **Nginx-served frontend** with a **Python backend**.

Publications

- **Autonomous Wheeled Robot for Intelligent Supermarket Restocking** | LINK — Jun 2025
  IEEE 2nd International Conference on Trends in Engineering Systems and Technologies (ICTEST)

Certifications

- **IBM RAG and Agentic AI** Professional Certificate | CERTIFICATE — Dec 2025
- **AWS Generative AI for Developers** Professional Certificate | CERTIFICATE — Oct 2025
- **IBM AI Engineering** Professional Certificate | CERTIFICATE — Jul 2025
- **PyTorch for deep learning** Professional Certificate | CERTIFICATE — Jun 2025

Languages

- **German:** B1 (Currently pursuing B2)
- **English:** C1